Chapter 8 — Design Patterns: Introduction & Creational Patterns (Study Edition)

> **Study Edition.** A shorter version of Chapter 8 with every topic kept and the padding removed. The full chapter (README.md) has the complete code, full outputs and more follow-ups.
>
> **What you will learn.** What a design pattern is, where the idea came from, all 23 GoF patterns, the OO principles behind them, how to pick (and not pick) a pattern, anti-patterns, and the creational patterns in depth: Singleton, Simple Factory (idiom), Factory Method, Abstract Factory, Builder, Prototype and the bonus Object Pool.
>
> **Code.** `chapters/08-patterns-creational/code/lld/ch08/<pattern>/`, one `Main` per package (Java 21: `javac --release 21`, then `java lld.ch08.<pattern>.Main`).

Part A — Design patterns: the big picture

8.1 What is a design pattern?

What it is

A **design pattern** is a named, reusable description of a good solution to a recurring design problem. It is not a library or paste-in code — more like a *recipe* you adapt. (Analogy: builders learned over hundreds of houses where the front door works best; writing that down, with reasons and costs, is a pattern.)

- **Intent** — the one-sentence purpose ("ensure a class has only one instance").
- **Participants** — the roles that take part (*Creator*, *Product*, *Client*).

The problem it solves

Without patterns every team re-solves the same problems in slightly different, slightly buggy ways. Patterns give proven solutions, readable code, built-in good principles (loose coupling, open/closed), adaptability and faster decisions — and a **shared vocabulary** ("use an Observer" replaces a paragraph), the biggest interview win.

How it works

Every GoF pattern has four parts: **Name**, **Problem** (when to apply, forces in tension), **Solution** (participants and collaborations — a template, not code), **Consequences** (costs and benefits — a pattern without trade-offs is half a pattern). You adapt the template to your language and problem.

When not to use it

Not every problem needs a pattern. Forcing one adds classes and indirection — worse than none (see 8.6).

TL;DR

Pattern = **name + problem + solution template + consequences**. Value: vocabulary and proven trade-offs. Risk: overuse.

8.2 A short history: from buildings to the "Gang of Four"

Year	Event
1977	Architect **Christopher Alexander**, *A Pattern Language* — named solutions for towns and buildings
1987	**Kent Beck** and **Ward Cunningham** apply pattern languages to Smalltalk UI design
1994	**Gamma, Helm, Johnson, Vlissides** — the **Gang of Four (GoF)** — publish *Design Patterns*: **23 patterns** in three categories
2004	*Head First Design Patterns* — the same patterns in Java, OO principles front and centre

The GoF's two central principles:

1. **Program to an interface, not an implementation.**
2. **Favour object composition over class inheritance** ("has-a" black-box reuse over "is-a").

Creational patterns make principle 1 practical: if code must not name concrete classes, *someone* still has to create the objects.

8.3 The three categories and all 23 GoF patterns

What it is

Category	In one line
Creational	Control creation so the system doesn't depend on concrete classes or construction details
Structural	Compose classes/objects into larger, flexible structures
Behavioral	Assign responsibilities and design communication

Second GoF axis, **scope**: **class patterns** use inheritance, fixed at compile time (Factory Method, class Adapter, Template Method, Interpreter); **object patterns** use composition, changeable at run time (the rest).

The full catalogue

#	Pattern	Category	Intent
1	**Abstract Factory**	Creational	Create *families* of related objects without naming concrete classes
2	**Builder**	Creational	Build a complex object step by step, separate from its representation
3	**Factory Method**	Creational	Subclasses decide which class to instantiate via an overridable method
4	**Prototype**	Creational	Create objects by copying a configured instance
5	**Singleton**	Creational	Exactly one instance with a global access point
6	**Adapter** (Wrapper)	Structural	Convert one interface into another
7	**Bridge**	Structural	Split abstraction from implementation so both vary
8	**Composite**	Structural	Treat single objects and trees uniformly
9	**Decorator**	Structural	Add responsibilities dynamically by wrapping
10	**Facade**	Structural	Simple entry point to a complex subsystem
11	**Flyweight**	Structural	Share intrinsic state across huge numbers of objects
12	**Proxy**	Structural	Stand-in controlling access to another object
13	**Chain of Responsibility**	Behavioral	Pass a request along handlers until one handles it
14	**Command**	Behavioral	Request as an object — queue, log, undo
15	**Interpreter**	Behavioral	Grammar as classes; evaluate sentences
16	**Iterator**	Behavioral	Traverse without exposing storage
17	**Mediator**	Behavioral	Centralise communication between objects
18	**Memento**	Behavioral	Capture/restore state without breaking encapsulation
19	**Observer**	Behavioral	Notify dependents when one object changes
20	**State**	Behavioral	Behaviour changes with internal state
21	**Strategy**	Behavioral	Interchangeable algorithms behind one interface
22	**Template Method**	Behavioral	Base class fixes the skeleton, subclasses fill steps
23	**Visitor**	Behavioral	Add operations without changing the classes

Creational → this chapter, structural → Ch. 9, behavioral → Ch. 10–11. **5 + 7 + 11 = 23.** (Some course lists show ten behavioral patterns, omitting the rarely used Interpreter.) **Not GoF but common:** Simple Factory (idiom), Object Pool, Dependency Injection, Null Object, Repository, MVC (a "compound pattern" of Observer + Strategy + Composite).

Self-check

- **Q:** Which creational pattern is a *class* pattern? **A:** Factory Method — subclasses override the creation method.
- **Q:** How many behavioral GoF patterns? **A:** Eleven.

8.4 The OO principles behind every pattern

Principle	Patterns applying it
Encapsulate what varies	Factory Method, Strategy, State, Builder
Composition over inheritance	Abstract Factory (client *holds* a factory), Strategy, Decorator
Program to interfaces	Every creational pattern returns an interface
Loose coupling	Observer, Mediator, factories
Open–Closed	Factory Method, Decorator
Dependency Inversion	All factories — the client never calls `new ConcreteThing()`
Least Knowledge	Facade
Hollywood Principle	Template Method, Factory Method, frameworks/IoC
Single Responsibility	Builder, Iterator

Why creational patterns exist: DIP says don't depend on concrete classes, but `new PdfDocument()` *is* such a dependency — so creational patterns move that `new` somewhere controlled (a factory, subclass, builder, prototype or container).

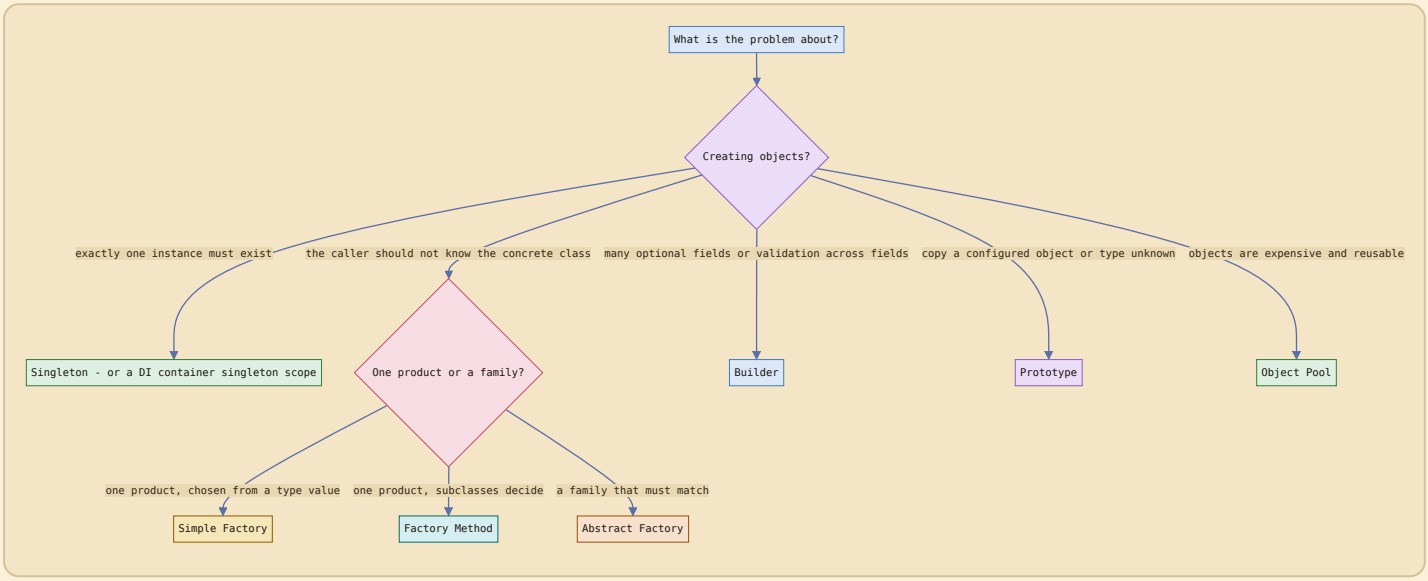
8.5 How to pick a pattern

The problem it solves

Learning 23 patterns and hunting for places to use them is backwards. Start from a **signal** in the requirements.

How it works

1. **Name the problem** in plain words ("pricing will change", "exactly one of these", "ten-argument constructor").
2. **Decide the category** — creating, structuring or behaving.
3. **Match the signal** to a pattern.
4. **Check the cost** — is there something simpler (enum, lambda, constructor, DI container)?
5. **Say what you rejected and why** — worth as much as the pick.



(For fitting objects together: Adapter, Decorator, Facade; for behaviour: Strategy, Observer, State.)

The six patterns that come up in almost every LLD interview

Learn each with its **signal** — forcing a pattern where it doesn't fit costs more than not knowing it.

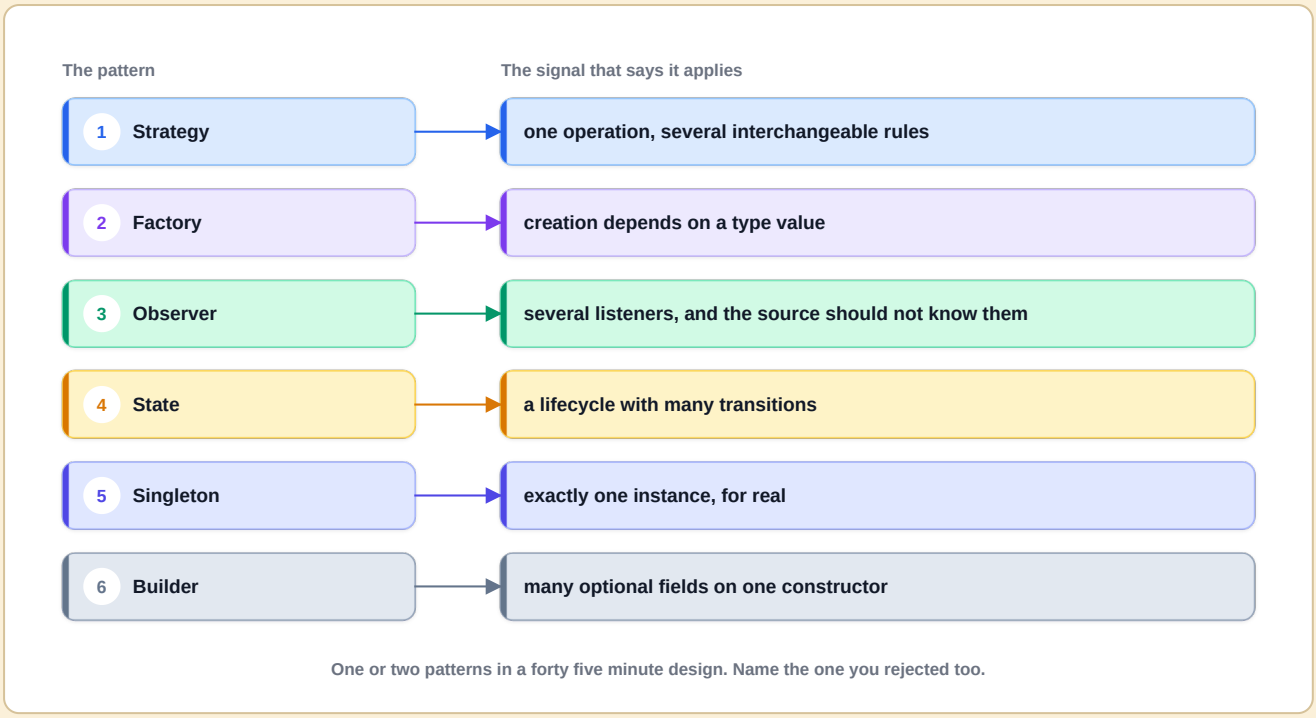


Figure — the six interview patterns and the signal that tells you each one applies (course diagram)

Pattern	Signal	Use when	Don't use when
Strategy	One operation, several swappable versions chosen at run time	Several interchangeable rules	Only one rule, no hint of a second
Factory	Something must be created; the caller shouldn't care which class	Class depends on a type value	Only one concrete class
Observer	One object changes, several others must react unknown to it	Several listeners	A single listener already referenced
State	Behaviour depends on status; allowed moves are fixed	Many states and transitions	3–4 states handled by one guarded method
Singleton	Genuinely one instance	Must be exactly one	You just want a global handle
Builder	Many optional fields; constructor keeps growing	Lots of optional fields	A few required fields

In a parking-lot / library design: **Strategy** — a `ParkingRate` interface (`HourlyRate`, `WeekendRate`, `MonthlyPass`), the answer to "what if pricing changes?"; **Factory** — type value → `Vehicle`, keep it lean (a creation method is often enough in 45 min); **Observer** — spot as subject, boards subscribe (but one reserved-book member needs only a direct call); **State** — ticket active → paid → closed (with 3–4 states an `enum` + guard is clearer); **Singleton** — one `ParkingLot`, raise thread safety and testability yourself; **Builder** — a `BookItem` past ~6 arguments.

Rules of thumb: one or two patterns per 45-minute design, placed where their signal appears; **compare out loud** — "Strategy for rates because a second tariff is coming; no State for the ticket because one guard handles three statuses."

8.6 Anti-patterns: when patterns go wrong

What it is

An **anti-pattern** is a common "solution" that looks reasonable but reliably causes more problems than it solves.

Anti-pattern	Looks like	Fix
God Object (Blob)	`SystemManager` with 80 methods — violates SRP, untestable	Extract Class (Ch. 7); move behaviour to data owners
Golden Hammer	Favourite tool/pattern applied everywhere	Start from the problem's signal (8.5)
Pattern overuse ("patternitis")	Factories for one class, interfaces "just in case" — **Speculative Generality**	YAGNI: add the pattern when the *second* variant arrives
Singletonitis	Singletons as globals — hidden deps, order-dependent tests	Dependency injection (8.8)
Cargo-cult patterns	Structure without intent (a "Builder" that validates nothing)	Know the property the pattern gives you, or drop it

TL;DR

Patterns are a vocabulary, not a checklist. God Object comes from ignoring design; Golden Hammer and overuse from over-applying it.

Part B — Creational patterns

8.7 Creational patterns at a glance

What it is

Creational patterns control *how objects are created*, hiding *which* class is instantiated, *how* it is assembled, and sometimes *whether* a new object is made. The system becomes independent of creation, hides creation complexity, and stays flexible about what/who/how/when.

Pattern	Intent
Singleton	Exactly one instance, one global way to reach it
Factory Method	Base class defines a creation method; subclasses choose the class
Abstract Factory	One factory object creates a matching *family*
Builder	Assemble one complex object step by step
Prototype	Create by cloning a prototypical instance
Simple Factory (idiom)	One method with a switch/registry picks the class
Object Pool (bonus)	Reuse a bounded set of expensive objects

8.8 Singleton

What it is

Singleton restricts a class to **exactly one instance** (per process / class loader) with **one well-known access point** — like a country's one government. **Eager** = create at class initialisation; **lazy** = on first request.

GoF intent and participants. Intent: never more than one instance; a well-known way to reach it. One participant, **Singleton**: a class-level `getInstance()` operation, usually also creating the instance. Clients just call it.

The problem it solves

Some things are one per process: settings loaded from a file, a connection pool, a metrics registry. If billing, reporting and the mailer each build their own `ConfigStore`, the file is parsed three times and one service's change never reaches the others. A plain global has two problems: tests share it (results depend on test order), and someone must assign it before use (wrong start-up order → `null`). Singleton makes **the class decide how many instances exist and create the one it allows**, on demand.

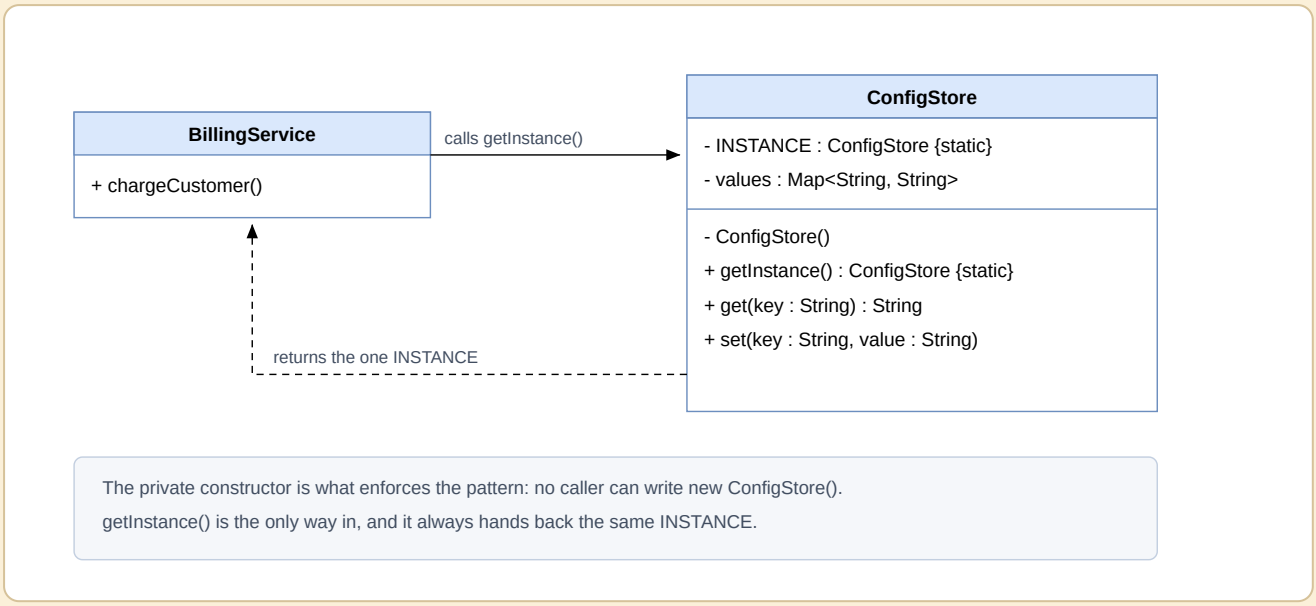
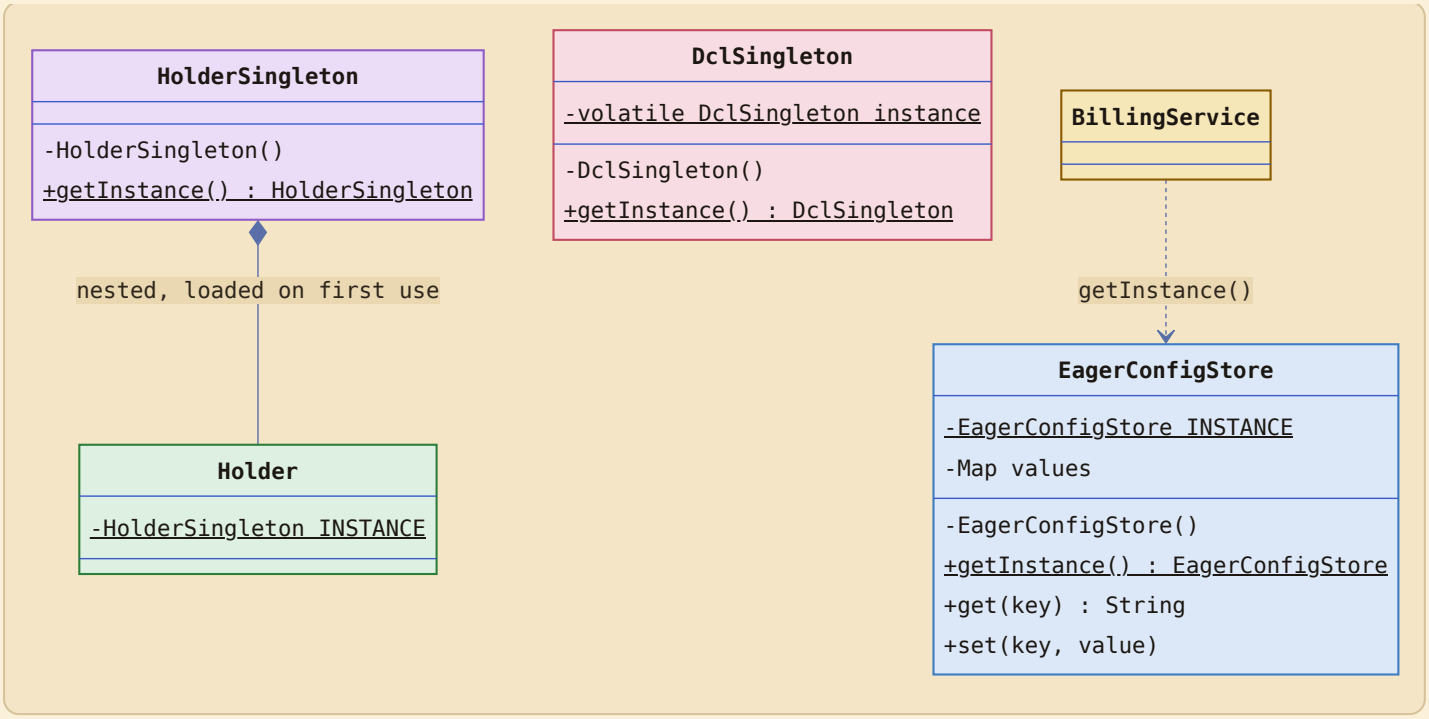


Figure — ConfigStore with a private static INSTANCE, a private constructor and a static getInstance(); BillingService only calls getInstance() (course diagram)

How it works

1. `private` `constructor` — the whole enforcement mechanism.
2. The instance in a `private static` `field` (owned by the class).
3. A `public static getInstance()` (static: callers have no instance yet).
4. Return the instance, creating it (safely) if needed.
5. Ordinary instance methods for the real work.



The six ways to write a Singleton in Java

#	Variant	Lazy?	Thread-safe?	Per-call cost	Survives serialization / reflection?	Verdict
1	**Eager** `static final`	No	Yes — class init runs once	None	No / No	Default when construction is cheap
2	**Naive lazy** `if (instance == null)`	Yes	**No**	None	No / No	Never in multi-threaded code
3	`synchronized` **getInstance()**	Yes	Yes	Lock on *every* call	No / No	Correct but slow under contention
4	**Double-checked locking** + `volatile`	Yes	Yes (Java 5+)	One volatile read	No / No	When creation needs run-time input
5	**Bill Pugh holder**	Yes	Yes — class loading	None	No / No	Best lazy option in plain Java
6	`enum`	No	Yes	None	**Yes / Yes**	Shortest, most robust (*Effective Java*)

- **Eager:** set during class *initialisation* (first active use, not JVM start); JLS §12.4.2 guarantees once-only, safely published. Cost: built even if the class is touched for another reason.
- **Naive lazy:** *check-then-act* — two threads both see `null` and both construct. Passes single-threaded tests; our 8-thread demo built **8** objects.
- `synchronized`: fixes the race but locks forever for a first-call problem.
- **DCL:** check without the lock, lock, **check again**. The field **must be** `volatile`: object creation is allocate → construct → store, and without `volatile` another thread may see the reference before the constructor's writes (a half-built object). `volatile` gives a **happens-before** edge. Copying to a local reads the volatile once on the fast path.
- **Holder:** `private static final class Holder { static final X INSTANCE = new X(); }` and `getInstance()` returns `Holder.INSTANCE`; `Holder` isn't initialised until first touched — lazy, thread-safe, **no locks, no** `volatile`.
- **Enum** (`enum EnumConfigStore { INSTANCE; ... }`): one instance per constant, reflective creation refused, serialization special-cased. Can't extend a class; not truly on-demand.

Which to pick: default **eager**; **holder** once you've *measured* a start-up cost; **DCL** when creation needs a run-time value (and always `volatile`); **enum** for free serialization/reflection safety.

```
/** 4. Double-checked locking: lock only on the first calls. volatile is mandatory. */
final class DclSingleton {
    static final AtomicInteger CONSTRUCTED = new AtomicInteger();
    private static volatile DclSingleton instance;
    // ... private constructor ...
    static DclSingleton getInstance() {
        DclSingleton local = instance;           // one volatile read on the fast path
        if (local == null) {                     // 1st check, no lock
            synchronized (DclSingleton.class) {
                local = instance;
                if (local == null) {             // 2nd check, under the lock
                    local = new DclSingleton();
                    instance = local;            // volatile write publishes a fully built object
                }
            }
        }
        return local;
    }
}
```

Breaking a Singleton — and the fixes

Attack	How	Fix
Reflection	`setAccessible(true).newInstance()` calls the private constructor	Throw from the constructor if `INSTANCE` exists (reliable for eager/holder, racy for lazy), or **enum**
Serialization	`readObject()` builds a new object without the constructor	`private Object readResolve() { return INSTANCE; }` (+ `transient` fields), or **enum**
Cloning	A `Cloneable` ancestor exposing `clone()`	Override `clone()` to throw (not an issue if no ancestor is `Cloneable` — `Object.clone()` is protected)
Multiple class loaders	Each loader has its own copy	Load from a shared parent loader, or let a container own it
Multiple processes	One per JVM — four servers, four instances	External coordinator (DB, Redis, ZooKeeper) for system-wide uniqueness

`HardenedSingleton` adds all three defences (constructor guard, `readResolve`, throwing `clone`). Output: the naive lazy version under 8 threads — `constructor ran 8 time(s)` (varies; anything above 1 is the bug), DCL — `1`; the hardened and enum versions print `reflection -> blocked` and `serialization -> same? true`.

Subclassing a Singleton

GoF: possible with a **`protected`** constructor, where `getInstance()` picks the subclass **once, at run time** — `AppLogger` chooses `PlainLogger` or `JsonLogger` from a system property. With a `private` constructor it can't be subclassed.

The dependency-injection alternative

Often the best singleton is **not a Singleton class**: a normal class, **one** instance created at the **composition root**, passed in through constructors. Dependencies become visible; tests pass a fake (`new BillingService(testConfig)`). Spring beans are **singleton-scoped by default** — one per bean definition **per container**, not per JVM, and nothing stops `new`. Spring's **`prototype` scope** creates a fresh instance per lookup.

Java implementation

The package also has `EagerConfigStore`, `EnumConfigStore`, `BreakableSingleton`/`HardenedSingleton`, `AppLogger` and `DiAlternative`.

Full runnable code: code/lld/ch08/singleton/ (see the Full Edition).

Trade-offs

Pros	Cons
Controlled access — one instance, one documented door	**Global state** — shared mutable state anyone can change
You choose when — eager or lazy	**Awkward to test** — needs a reset/injection hook that ships to production
Fits shared resources — pool, cache, device handle	**Invisible dependencies** — `getInstance()` inside isn't in the constructor
Setup paid once per process	**Uniqueness ends at the process** (class loader)
Subclassable if designed in (`protected`)	**Hard to undo**; mixes lifecycle management with real work (SRP)

Often confused with: a **static utility class** (not an object — can't implement an interface or be mocked); **DI singleton scope** (same result, better mechanism); **Multiton** (`Logger.getLogger(name)` — one per key); Prototype (opposite intent). Elsewhere: Python module object, Go `sync.Once`, C++ Meyers Singleton.

When not to use it

- **The motive is convenience** — a parameter delivers access just as well.
- **Mutable state tests touch** — tests pass alone, fail in the suite.
- **A framework manages lifecycles** — use singleton scope.
- **Multiple processes** — four servers = four "singletons"; using one for distributed locks or unique IDs is a **correctness bug**.

Common mistakes

Naive lazy in multi-threaded code; DCL without `volatile`; `Serializable` without `readResolve()`; mutable state without thread-safe structures (`ConfigStore` uses `ConcurrentHashMap`); a `public static reset()` "for tests"; assuming one per system rather than per JVM/class loader.

Real-world examples

- **JDK:** `Runtime.getRuntime()` (textbook), `Desktop.getDesktop()`; `Collections.emptyList()` (singleton-like shared constant); `Logger.getLogger(name)` (multiton).
- **Not singletons despite `getInstance()`:** `Calendar.getInstance()`, `NumberFormat.getInstance()`, `Cipher.getInstance()` return **new objects** — static factory methods (8.9).
- Spring singleton beans; connection pools, metrics registries; *Head First*'s chocolate boiler.

In the interview

> **30-second answer:** "Singleton guarantees one instance per JVM (strictly, per class loader) with a single access point: private constructor, static accessor. I'd use eager initialisation — the class loader makes it thread-safe for free — or the holder idiom if construction is expensive; an enum if it must survive serialization or reflection. But it's global state with hidden dependencies, so in a Spring app I'd make it a singleton-scoped bean and inject it."

- *Q: Why does DCL need `volatile`?* — A: Otherwise the reference can be published before the constructor's writes are visible; `volatile` gives a happens-before edge.
- *Q: Unique IDs across 10 servers?* — A: Not a Singleton (one per JVM). Use a DB sequence, Snowflake-style IDs (timestamp + machine id + counter) or UUIDs.

Self-check

- **Q:** Which keyword enforces the pattern? **A:** `private` on the constructor.

- **Q:** Lazy, lock-free, thread-safe variant? **A:** The holder idiom.
- **Q:** What does `readResolve` do? **A:** Replaces the deserialized object with the real instance.

TL;DR

Private constructor + static accessor = one per JVM. Prefer **eager**, then **holder**, then **enum**; **DCL needs** `volatile`; never naive lazy. Know the ways to break it. Mention the costs; DI singleton scope is usually better.

8.9 Simple Factory (an idiom, not a GoF pattern)

What it is

One class (usually one static method) takes a type value and returns the matching object as a common interface. Very common, often right, but **not** GoF — *Head First* calls it an idiom. Analogy: a vending machine — press "B3", get the snack.

The problem it solves

The `switch` turning `"sms"` into `new SmsNotification()` gets copied everywhere. A simple factory **centralises** the decision — it doesn't **remove** it: adding a channel still edits the factory. A registry softens this (types are *registered* at start-up).

How it works

1. Product interface (`Notification`). 2. Concrete classes. 3. One `create(type)` — an exhaustive `switch` over an enum, or a `Map<String, Supplier<Notification>>` registry. 4. Callers hold the interface, never `new`. 5. Reject unknown types loudly.

Java implementation

```java
/** Simple Factory (an idiom, not a GoF pattern): one static method owns the "which class?" decision. */
public final class NotificationFactory {

    public enum Channel { EMAIL, SMS, PUSH }

    private NotificationFactory() { }

    /** Version 1: a switch. Adding a channel means editing this method (and the enum). */
    public static Notification create(Channel channel) {
        return switch (channel) {                    // exhaustive switch: the compiler checks every enum value
            case EMAIL -> new EmailNotification();
            case SMS   -> new SmsNotification();
            case PUSH  -> new PushNotification();
        };
    }

    /** Version 2: a registry of suppliers. New types are registered, not coded into a switch. */
    private static final Map<String, Supplier<? extends Notification>> REGISTRY = new ConcurrentHashMap<>(Map.of(
            "email", EmailNotification::new,
            "sms", SmsNotification::new,
            "push", PushNotification::new));

    public static void register(String type, Supplier<? extends Notification> supplier) {
        REGISTRY.put(type.toLowerCase(Locale.ROOT), supplier);
    }

    public static Notification create(String type) {
        Supplier<? extends Notification> s = REGISTRY.get(type.toLowerCase(Locale.ROOT));
        if (s == null) throw new IllegalArgumentException("Unknown notification type: " + type);
        return s.get();
    }
}
```

- No `default` in the enum switch — the compiler flags an unhandled constant.
- The registry maps names to constructor references; `register("whatsapp", ...)` adds a channel without editing the factory (a lambda works because `Notification` has one abstract method).
- `Locale.ROOT` avoids the Turkish dotless-"i" lower-casing surprise.

Full runnable code: code/lld/ch08/simplefactory/ (see the Full Edition).

Trade-offs

Simple Factory	Factory Method
One method, usually a switch	Decision spread across subclasses
New type = edit the factory	New type = add a class
No inheritance	Creator subclass per product
Not GoF	GoF

Short, stable, config-selected type list → Simple Factory is perfectly respectable. Other developers/modules adding types without editing your class → Factory Method.

When not to use it

Only **one** concrete class (call the constructor); objects differ only in **data** (one class with a field).

Common mistakes

Calling a switch-based factory a "Factory Method"; a `default` that returns `null` or a silent default; letting the factory grow into a God class.

Real-world examples

Static factory methods (*Effective Java* item 1) — not GoF Factory Method, usually not singletons: `Calendar.getInstance()` (new object per call), `EnumSet.noneOf` (hidden `RegularEnumSet`/`JumboEnumSet`), `List.of`, `Integer.valueOf`, `Executors.newFixedThreadPool`; Spring `@Bean` methods.

In the interview

> **30-second answer:** "A simple factory is one method that maps a type value to a concrete class and returns it as an interface. It centralises the switch so callers don't depend on concrete classes. It's an idiom, not GoF — adding a type still edits it. For new types without editing, I'd use a registry of suppliers or Factory Method."

- *Q: Is it Open–Closed?* — A: Not strictly; a registry makes it open via registration.
- *Q: Static factory vs constructor?* — A: Names, caching (`Integer.valueOf`), can return subtypes.

Self-check

- **Q:** What does it centralise but not remove? **A:** The switch — knowledge of every concrete type.
- **Q:** Why no `default` in the enum switch? **A:** So the compiler flags unhandled constants.

TL;DR

One `create(type)` returning an interface. Great for small stable sets. Not GoF. `Supplier` registry = extensible variant.

8.10 Factory Method

What it is

A base class defines a creation method but **subclasses decide the concrete class**; the base class uses the product without naming its type. Analogy: a franchise manual — take order, *prepare the dish* (each branch's own), serve, bill (*Head First*'s `PizzaStore.orderPizza()` with regional `createPizza()`).

GoF intent and participants. Intent: declare an operation that creates an object but let subclasses choose the class — instantiation is deferred to them. **Product** (`Document`), **ConcreteProduct** (`PdfDocument`), **Creator** (`ReportGenerator` — declares the factory method and *calls* it from its own operations), **ConcreteCreator** (`PdfReportGenerator` — overrides it).

> **Not a "static factory method".** `List.of`, `Calendar.getInstance` and Simple Factories are **one static method that decides** — not overridable, no Creator hierarchy. GoF Factory Method is an **instance method overridden in subclasses** and called by the base class. No inheritance + overriding → not this pattern.

The problem it solves

`new PdfDocument()` inside `ReportGenerator` hard-wires PDF. The generator gathers rows, **creates a document**, renders, announces — only creation depends on format. A `switch` works until a second and third method need the same switch. Factory Method leaves a deliberate **gap**: the full procedure in the base class calls an abstract method each subclass fills.

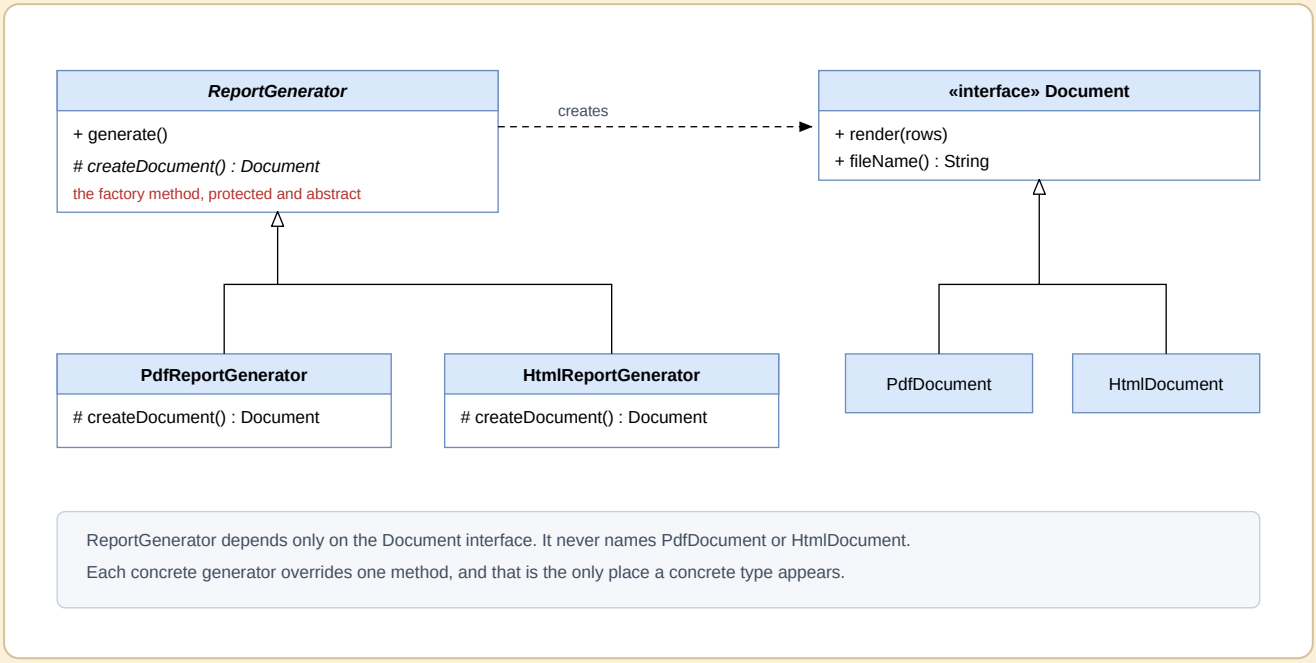
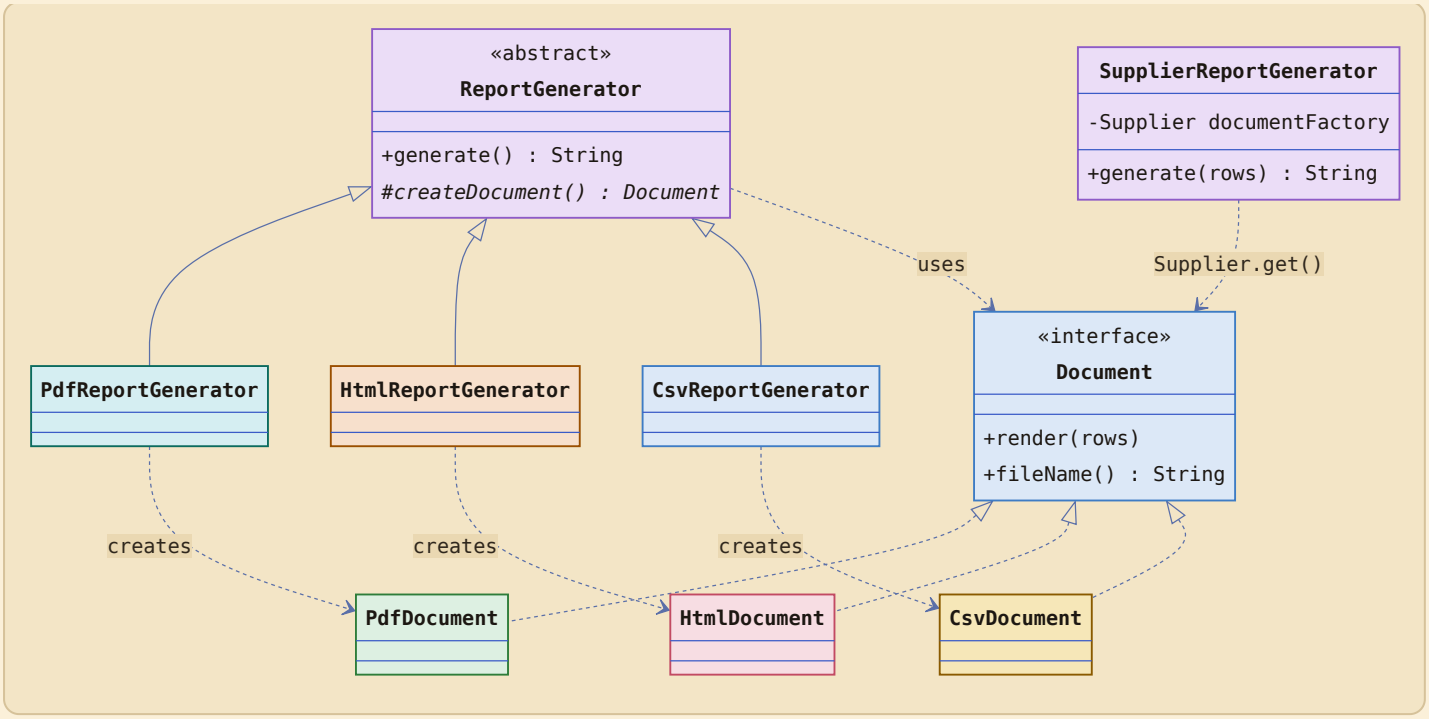


Figure — ReportGenerator declares the protected factory method createDocument(); each concrete generator overrides it to return its own Document (course diagram)

How it works

1. Product interface (`Document`: `render(rows)`, `fileName()`). 2. Concrete products. 3. Abstract creator with an **abstract** **protected** `createDocument()` (an extension point, not API). 4. Shared logic in the creator (`generate()`) calling it. 5. One concrete creator per product — the *only* place a concrete type is named. 6. The client picks a creator once, then uses `ReportGenerator` and `Document`.



Java implementation

```java
/** Creator: writes the whole procedure once and leaves one hole - the factory method. */
public abstract class ReportGenerator {

    /** The factory method. Protected: a hook for subclasses, not public API. */
    protected abstract Document createDocument();

    /** The template that USES the factory method. It never names a concrete document. */
    public final String generate() {
        System.out.println("  Collecting rows for the monthly sales report...");
        List<String> rows = List.of("north,120", "south,95", "east,140");
        Document document = createDocument();
        document.render(rows);
        System.out.println("  Report ready: " + document.fileName());
        return document.fileName();
    }
}

final class PdfReportGenerator extends ReportGenerator {
    @Override protected Document createDocument() { return new PdfDocument(); }
}
// ... HtmlReportGenerator, CsvReportGenerator (CSV added later - no existing class edited) ...

/** The modern, composition-based variant: pass the "factory method" in as a Supplier. */
public final class SupplierReportGenerator {
    private final Supplier<? extends Document> documentFactory;
    // ... constructor ...
    public String generate(List<String> rows) {
        Document d = documentFactory.get();
        d.render(rows);
        return d.fileName();
    }
}
```

`generate()` is `final` — subclasses change *what* is created, not the procedure. `CsvDocument` + `CsvReportGenerator` were added with no edits (Open–Closed). The `Supplier` variant composes better and a test can pass a lambda. In the output, every generator prints the same "Collecting rows…" line — that code exists once.

Full runnable code: code/lld/ch08/factorymethod/ (see the Full Edition).

Trade-offs

Pros	Cons
Extensible without edits — new product = new class	**More classes** — a creator per product
Common procedure in one place	**Indirection** — which subclass am I holding?
Test seam — subclass returns a fake	**Relies on inheritance** (one superclass)
Nobody holds the full type list	**Tempting too early** — all cost with one product

Often confused with: Simple Factory (one static switch); Abstract Factory (a *family* via a passed-in factory); **Template Method** (same hook mechanism but varies a *step* — `generate()` *is* one); Builder (many calls). In languages with first-class functions, passing a creation function is usually the better default.

When not to use it

One product type and no second coming (most frequent over-design); the base class has no shared work; types differ by data (use `Map<String, Supplier<Document>>`); you need matching objects (Abstract Factory).

Common mistakes

A **public** factory method (callers bypass the procedure); calling it from the base **constructor** (subclass fields not yet initialised); calling a switch-in-one-method a Factory Method.

Real-world examples

- **JDK:** `Collection.iterator()` — each collection (ConcreteCreator) returns its own `Iterator` (ConcreteProduct); `AbstractList.iterator()` built on abstract `get(int)`.
- **JDK XML:** `DocumentBuilderFactory.newDocumentBuilder()`, `TransformerFactory.newTransformer()` — factory methods inside an Abstract Factory (`newInstance()` itself is a *static* factory).
- Spring `FactoryBean<T>.getObject()`; Android `Fragment.onCreateView()`.

In the interview

> **30-second answer:** "Factory Method puts a creation hook in a base class. The base class writes the procedure once against a product interface; each subclass overrides the protected factory method to return its product. New types are new classes — Open–Closed. It pays off only with real shared logic in the base class; otherwise a constructor or simple factory is simpler. In modern Java I'd often pass a `Supplier`."

- *Q: vs Simple Factory?* — A: One static switch that knows every type vs subclasses overriding a hook the base calls. Raise this yourself — calling a switch a Factory Method is a classic slip.
- *Q: How do you test `generate()`?* — A: Subclass the creator (or pass a `Supplier`) returning a fake `Document`.

Self-check

- **Q:** The only place a concrete product is named? **A:** Each concrete creator's override.
- **Q:** What must the base class have for this to be worthwhile? **A:** Shared logic using the product.
- **Q:** Adding `CsvReportGenerator` shows which principle? **A:** Open–Closed.

TL;DR

Base class = procedure + abstract `protected createX()`; subclasses pick the type. One product, varied by inheritance. `Supplier` is the light modern form.

8.11 Abstract Factory

What it is

An interface for creating **families of related objects** that must be used together, without naming concrete classes. Choosing a family = choosing a factory. Analogy: a furniture catalogue — order from "Victorian" and every piece matches. (Course: one factory per OS theme; *Head First*: a regional pizza-ingredient factory.)

GoF intent and participants. Intent: one interface to create a set of related objects without naming their classes. **AbstractFactory** (`StorageFactory` — one create method per product kind), **ConcreteFactory** (`PostgresFactory`), **AbstractProduct** (`Connection`, `QueryBuilder`), **ConcreteProduct** (`PostgresQueryBuilder`), **Client** (`ReportService` — only abstract types). GoF notes one ConcreteFactory per family is typical, hence often a Singleton (passing it in is better).

The problem it solves

Some objects are only correct with partners from the same variant: Postgres quotes identifiers with `"`, MySQL with backticks; macOS buttons with macOS checkboxes; fake payments + fake mailer + fake clock, never a mix. Pairing `PostgresConnection` with `MySqlQueryBuilder` compiles and fails later at the database. Factory Method varies **one** product; Abstract Factory keeps a **set** consistent. *It doesn't help you choose the right combination — it removes the ability to build a wrong one.*

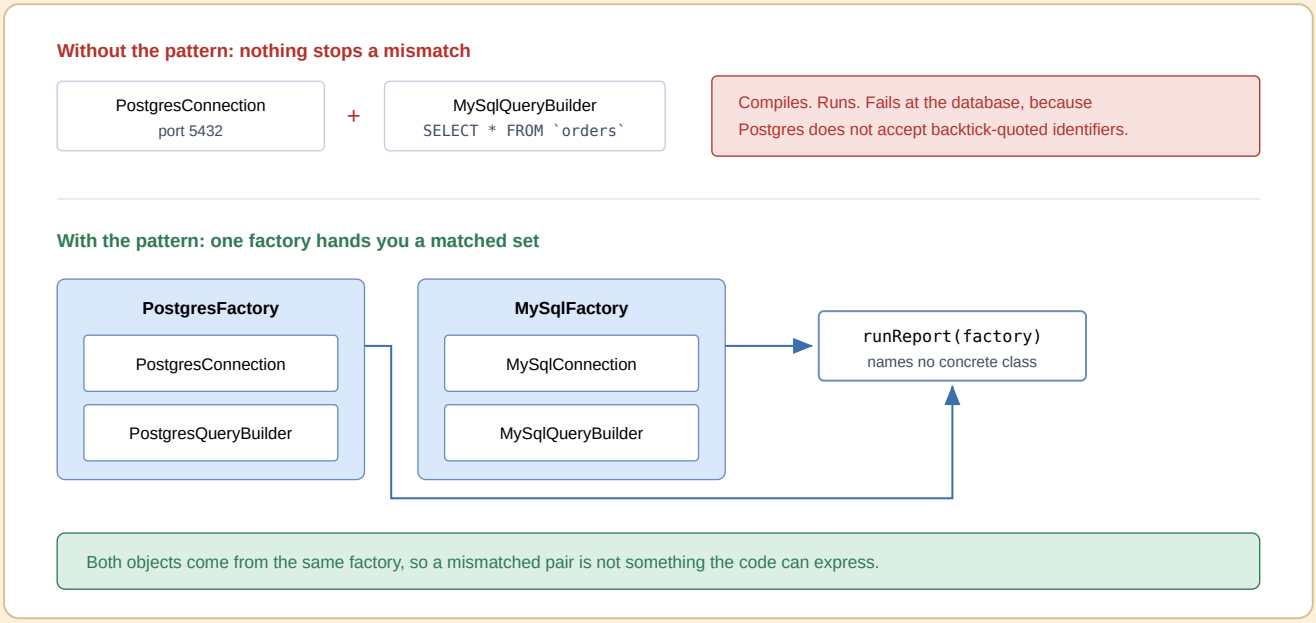
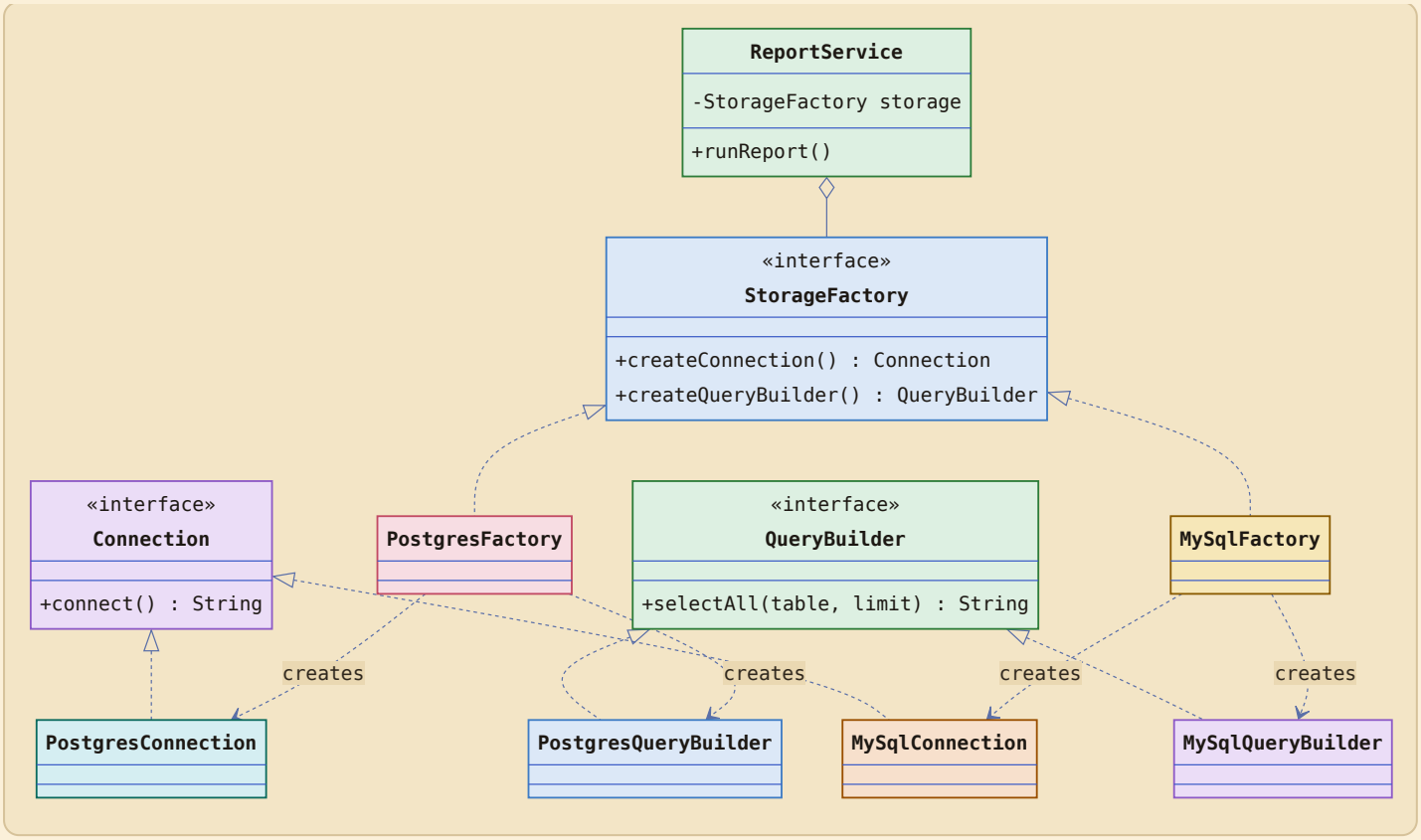


Figure — without the pattern a PostgresConnection can be paired with a MySqlQueryBuilder and fail at the database; with it, each factory hands out a matched set (course diagram)

How it works

1. Interface per **product kind**. 2. **Abstract factory** with one create method per kind. 3. Per variant, concrete products and a **concrete factory implementing every method** (covariant returns document the family). 4. The **client** receives a factory (constructor) and uses only abstract types. 5. Choose the factory **once at the composition root** (near `main`) and pass it down. (A correct Abstract Factory returns **all** products of its family — not one each.)



Java implementation

```java
/** Abstract factory: one creation method per product in the family. */
public interface StorageFactory {
    Connection createConnection();
    QueryBuilder createQueryBuilder();
}

/** Concrete factories implement EVERY method and return only their own family (covariant return types). */
final class PostgresFactory implements StorageFactory {
    @Override public PostgresConnection createConnection()     { return new PostgresConnection(); }
    @Override public PostgresQueryBuilder createQueryBuilder() { return new PostgresQueryBuilder(); }
}
// ... MySqlFactory, FakeStorageFactory ...

/** Client: holds only the abstract factory and abstract products. It cannot mix families. */
public final class ReportService {
    private final StorageFactory storage;

    public ReportService(StorageFactory storage) { this.storage = storage; }

    public void runReport() {
        Connection connection = storage.createConnection();
        QueryBuilder queries = storage.createQueryBuilder();
        System.out.println("  " + connection.connect());
        System.out.println("  " + queries.selectAll("orders", 10));
    }
}
```

`ReportService` has **no branch** on the database. `Main.chooseFactory(db)` — the composition root — holds the only `switch`. `FakeStorageFactory` swaps the **whole environment** for tests in one line.

Full runnable code: code/lld/ch08/abstractfactory/ (see the Full Edition).

Trade-offs

Pros	Cons
Wrong combinations can't be written	**Many types** — interface per kind + class per product per family
One switch for a whole environment (real↔fake, Postgres↔MySQL, themes)	**New product kinds are expensive** — edit every factory
Client never names a concrete class (DIP)	**Another layer to trace**
Test seam — one fake factory	**Pointless with one family**

The asymmetry: new **family** (`SqliteFactory`) = easy, Open–Closed; new **product kind** (`createMigrationRunner()`) = edit every factory.

Abstract Factory vs Factory Method

	Factory Method	Abstract Factory
Creates	**One** product	A **family**
Mechanism	**Inheritance** — subclass overrides	**Composition** — client holds a factory
Who chooses	Creator subclass	Factory passed in
Guarantees	New types without editing the creator	Products from one factory are compatible
New variant	New creator subclass + product	New factory + one product per kind
New product kind	n/a	Edit the interface and **every** factory
Example	`Collection.iterator()`	`DocumentBuilderFactory`, UI toolkits, DB dialects

Often confused with: Builder (one product, many steps); a DI container (same outcome via profiles — prefer it if you have one); Facade (simplifies, no consistency guarantee).

When not to use it

Objects that don't belong together (`createLogger()` + `createInvoice()` is a grab-bag); one variant only; a product-kind list that keeps growing (> 5–6); your DI container already does it (`@Profile`, Guice modules).

Common mistakes

A factory implementing only some methods or mixing families; choosing the factory deep in the call stack; making it a Singleton by reflex.

Real-world examples

- **JDK:** `DocumentBuilderFactory`, `TransformerFactory` (creates `Transformer` and `Templates`), `XPathFactory` — `newInstance()` locates an implementation whose products all match. JDBC `Connection` creates driver-specific `Statement`, `PreparedStatement`, `Blob`.
- UI toolkits (GoF's example), Hibernate `Dialect`, Spring `@Profile` bean sets.

In the interview

> **30-second answer:** "Abstract Factory is an interface with one create method per product in a family; each concrete factory returns a matched set. The client holds only the factory and abstract products, so it can't mix a Postgres connection with a MySQL query builder. I pick the factory once at the composition root. The cost: a new product kind means editing every factory."

- *Q: Add SQLite?* — A: `SqliteConnection`, `SqliteQueryBuilder`, `SqliteFactory`, one line in the composition root.

- *Q: Add a `MigrationRunner`?* — A: New method on the interface and every factory — the weak spot.

Self-check

- **Q:** What does it guarantee? **A:** Products from one factory are always the same family.
- **Q:** What is the composition root? **A:** The one place near start-up where concrete choices are made and passed down.
- **Q:** Easy vs hard change? **A:** New family easy; new product kind hard.

TL;DR

One factory → a matched family. Chosen once, passed down. Easy families, hard product kinds. Prefer DI profiles if you have a container.

8.12 Builder

What it is

Builder separates the *construction* of a complex object from the object. A builder collects values **one named step at a time** (a **fluent interface** — each returns `this`); `build()` produces the finished, usually **immutable**, product. Analogy: ordering at Subway, step by step.

	GoF Builder (1994)	**Bloch's Builder (*Effective Java* item 2)**
Targets	One construction process → **different representations** (RTF reader → ASCII, TeX, text widget)	Many optional parameters; wish for an immutable, validated object
Participants	**Builder** (abstract, one method per part), **ConcreteBuilder** (one representation, `getResult()`), **Director** (runs the steps), **Product**	Product + static nested `Builder`: fluent setters, validating `build()`; no Director or abstract Builder
What varies	Which ConcreteBuilder is plugged in	Only the values

Most real code is Bloch's; interviews may expect the GoF participant names too.

The problem it solves

`new HttpRequest(url, "GET", headers, null, 30, 3, true, false, null)` — which `null` is the body? Swap two booleans and it still compiles.

Approach	**What goes wrong**
Telescoping constructors	Overloads explode; still positional; same-typed params swap silently
JavaBeans (empty constructor + setters)	Half-built objects; can't be immutable; nowhere to validate cross-field rules

Builder: **named** values, **defaults**, **immutable** product, and `build()` as **one gate**. Our rule "a GET must not carry a body" (no defined meaning per RFC 9110 §9.3.1) spans two fields, so only `build()` can enforce it.

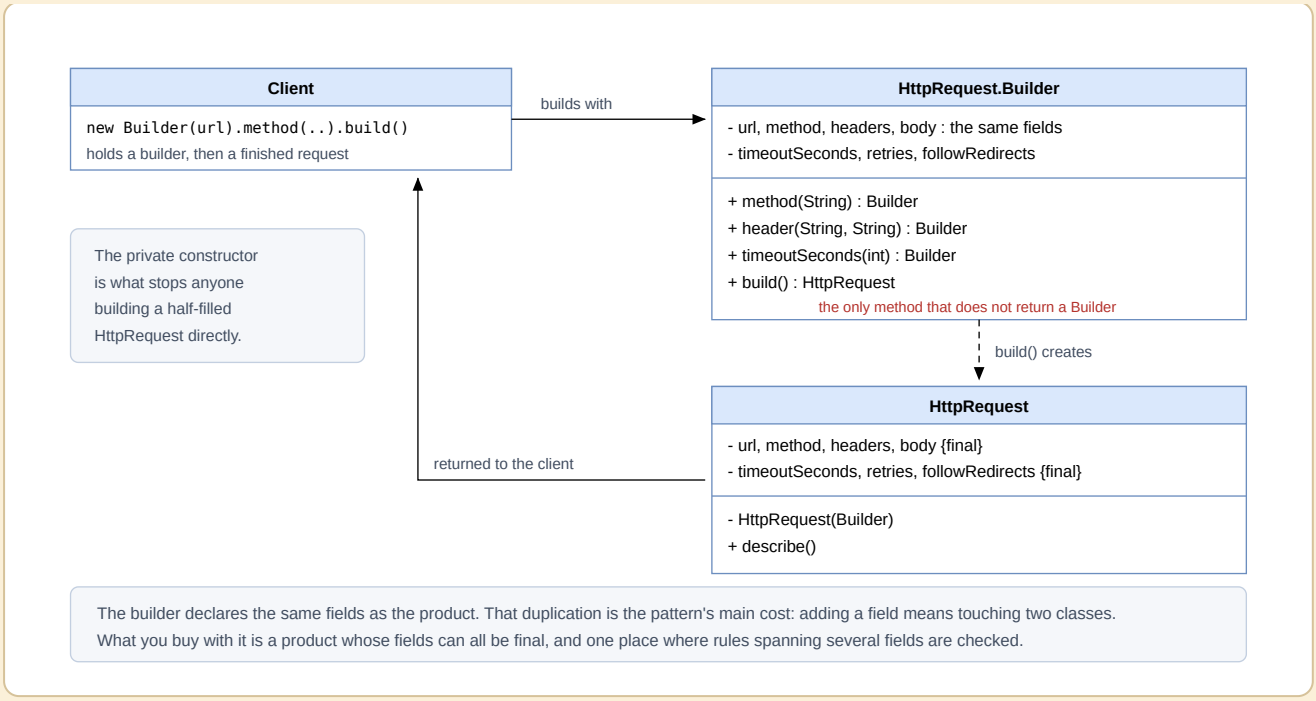
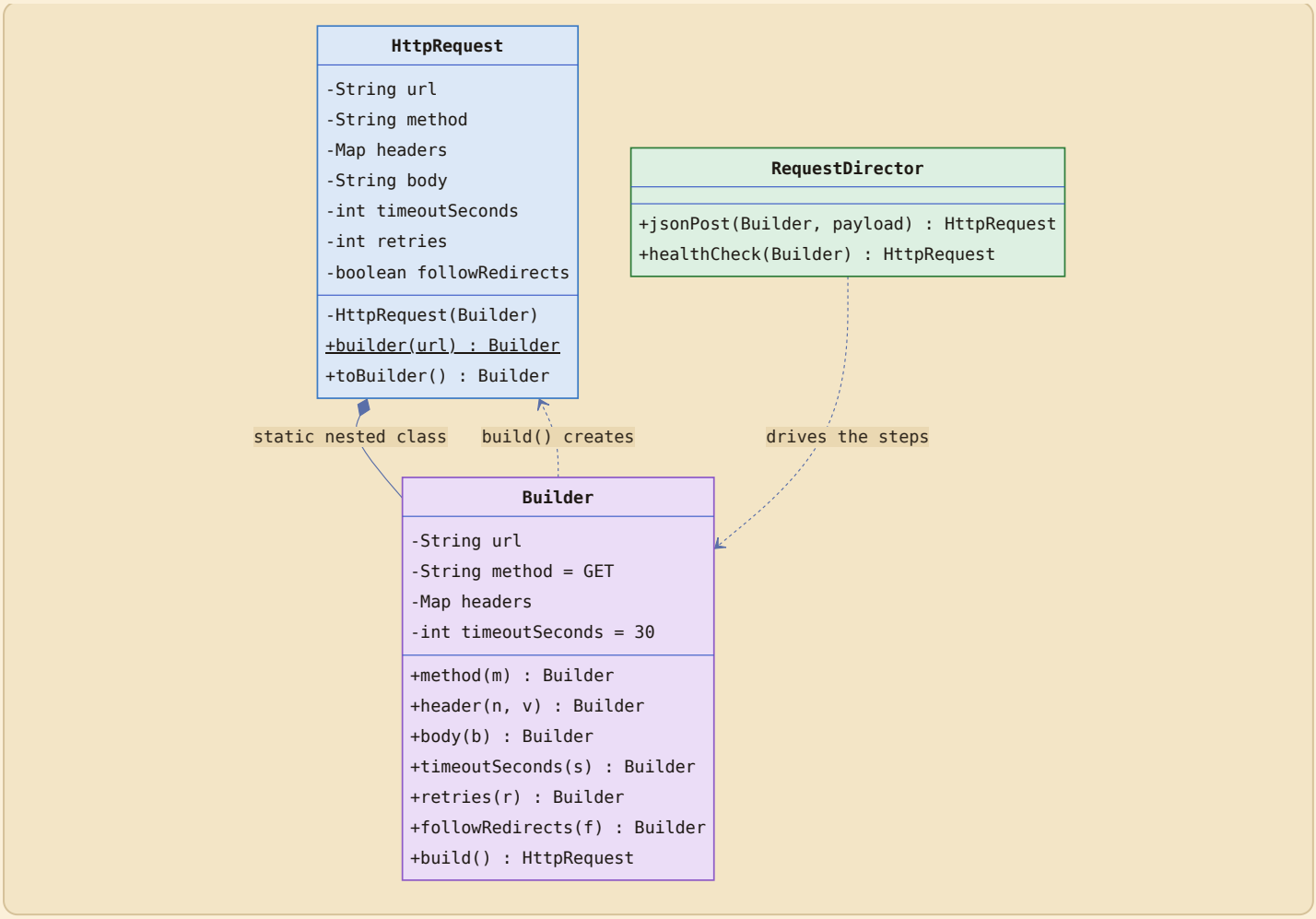


Figure — HttpRequest with final fields and a private constructor; HttpRequest.Builder with the same mutable fields, setters returning Builder, and build() returning HttpRequest (course diagram)

How it works

1. Product fields `final`, constructor `private` taking the builder. 2. `static` **nested** `Builder` with mutable fields and **defaults**. 3. **Required** values in the builder's constructor / `builder(url)`. 4. Named step methods that `return this`. 5. `build()`: validate cross-field rules, **copy mutable collections**, call the private constructor — the only method returning the product. 6. Optional: **Director**, `toBuilder()`, **staged builder**. Price: every field is written twice.



The Director — and why you rarely see one

The GoF **Director** knows the **order of steps** and runs them on whatever builder it gets — a home for a reusable recipe. Day to day, only *values* vary, so callers chain the calls themselves. **Test:** will the Director ever get a *different kind* of builder? If not, drop it. (Our `RequestDirector` only takes `HttpRequest.Builder` — it shows just the "named recipe" half.)

The staged (step) builder

Encode required/ordered steps in **types**: `Email.builder()` returns `ToStep` (only `to()`), then `SubjectStep` (only `subject()`), then `OptionalStep` with `build()`. `Email.builder().build()` **does not compile** — the answer to "how do you stop callers forgetting a required field?".

Lombok-style builders and records

- **Lombok** `@Builder` generates it all (`toBuilder = true`, `@Singular`) but does **not** validate by default. **Records** give the immutable product — validate in the compact constructor. **Immutability checklist:** final class/private constructor, `final` fields, no setters, defensive copies, no leaking getters.

Java implementation

```
// ... HttpRequest: final fields; private HttpRequest(Builder b) copies them, with headers = Map.copyOf(b.headers)
    /** The builder: same fields, mutable, with defaults. Every setter returns this. */
    public static final class Builder {
        private final String url;                                  // required -> constructor
        private String method = "GET";                             // optional -> defaults
        private final Map<String, String> headers = new LinkedHashMap<>();
        private String body;
        private int timeoutSeconds = 30;
        private int retries = 3;
        private boolean followRedirects = true;

        private Builder(String url) { this.url = Objects.requireNonNull(url, "url"); }

        public Builder method(String m)            { this.method = m; return this; }
        public Builder header(String n, String v)  { headers.put(n, v); return this; }
        public Builder body(String b)              { this.body = b; return this; }
        // ... timeoutSeconds, retries, followRedirects ...

        /** The single gate: cross-field validation happens here, before any product exists. */
        public HttpRequest build() {
            if (url.isBlank()) throw new IllegalStateException("url is required");
            if (method.equals("GET") && body != null) throw new IllegalStateException("a GET request cannot have a
body");
            if (retries < 0) throw new IllegalStateException("retries cannot be negative");
            if (timeoutSeconds <= 0) throw new IllegalStateException("timeout must be positive");
            return new HttpRequest(this);
        }
    }
}
```

- `TelescopingPizza` is the anti-example (five constructors; `("large", "thin", true, false, true)`).
- `Map.copyOf` matters: without it, two products built from one builder share one map, and a later header leaks into the earlier product. It's also unmodifiable, so `headers().put(...)` throws.
- `toBuilder()` gives "same as this, but…" copies; `RequestDirector.jsonPost/healthCheck` are named recipes; `Email` is the staged builder.

Full runnable code: code/lld/ch08/builder/ (see the Full Edition).

Trade-offs

Pros	Cons
Self-describing call sites	**Duplicated fields** — two edits per new field
Immutable products, thread-safe to share	**Forgettable** `build()`
Home for validation	**Overused** for 3-field objects
Free optional fields via defaults	**Reuse pitfall** — shared collections unless copied
Different representations (GoF Director)	One extra short-lived object (rarely measurable)

Often confused with: Factory Method (one call picks a type); Abstract Factory (several products at once); a **fluent interface** (a calling style — no separate product); Prototype (starts from a finished object). Named/default arguments (Python, Kotlin) usually replace builders.

When not to use it

A few required fields (the most common misuse); named/default arguments available; a builder that validates nothing for a mutable product; a mutable object living inside one method.

Common mistakes

Setters not returning the builder; no validation in `build()`; sharing mutable collections with products; a public product constructor (bypasses validation); a Director when the order never varies.

Real-world examples

- **JDK:** `StringBuilder`, `HttpClient.newBuilder()`, `HttpRequest.newBuilder()`, `Locale.Builder`, `Calendar.Builder`, `Stream.builder()`, `Thread.ofVirtual().name("w").start(task)`, `ProcessBuilder`.
- **Spring/libraries:** `UriComponentsBuilder`, `ResponseEntity.ok()...`, `WebClient.builder()`, OkHttp, Protobuf, Guava `ImmutableList.builder()`.

In the interview

> **30-second answer:** "With many optional parameters I use a Builder: required values in the builder's constructor, one named method per optional field returning the builder, and `build()` that validates cross-field rules and creates an immutable product via a private constructor. I'd mention the GoF Director but usually skip it since only values vary. For tiny objects or languages with named arguments, I wouldn't bother."

- *Q: How do you enforce required fields?* — A: Builder constructor, validation in `build()`, or a staged builder.
- *Q: Is the built object thread-safe?* — A: The immutable product is; don't share the builder across threads.

Self-check

- **Q:** The only builder method not returning the builder? **A:** `build()`.
- **Q:** What bug does copying the headers map prevent? **A:** Two products sharing and mutating one map.
- **Q:** When is a Director worth it? **A:** When the same recipe is reused, possibly with different builders.

TL;DR

Named steps + defaults + one validating `build()` → immutable product. Fixes telescoping constructors and half-built JavaBeans. Director optional; staged builder for compile-time required steps; Lombok/records cut boilerplate.

8.13 Prototype

What it is

Create objects by **copying an existing, configured object** via its own `copy()` — the caller needn't know the concrete class. Analogy: copying a contract template and changing the name and date.

GoF intent and participants. Intent: use a configured instance as the model and make new objects by copying it. **Prototype** (declares the copy operation — `Shape.copy()`), **ConcretePrototype** (`Circle`, `Rectangle`, `Group`; `NotificationTemplate` plays both roles), **Client** (asks a prototype to copy itself — `Main`, or `ShapeRegistry`, a *prototype manager*).

The problem it solves

1. **Expensive construction** — a `NotificationTemplate` parsed from disk; each welcome email differs only in subject and one channel. Re-parsing per recipient repeats the slowest work.
2. **Unknown concrete class** (the more important case) — you hold a `Shape` and need one more of the same; you can't write `new`. The object knows its own class, so it copies itself — no factory or `switch`, and the copy starts from correct values.

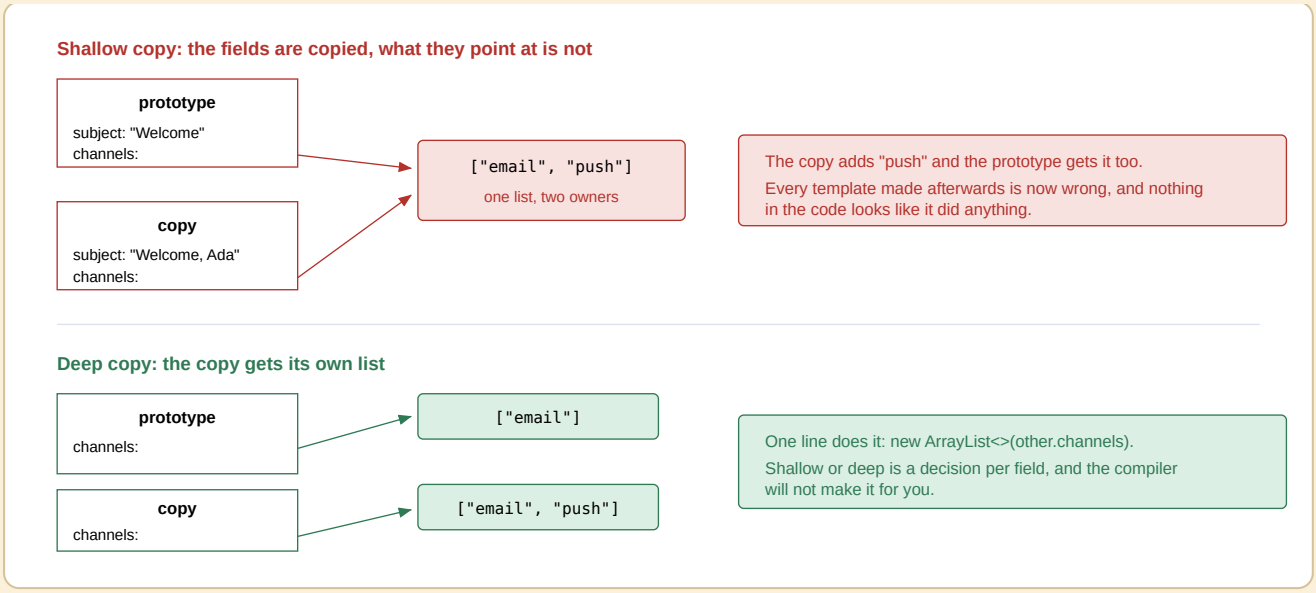
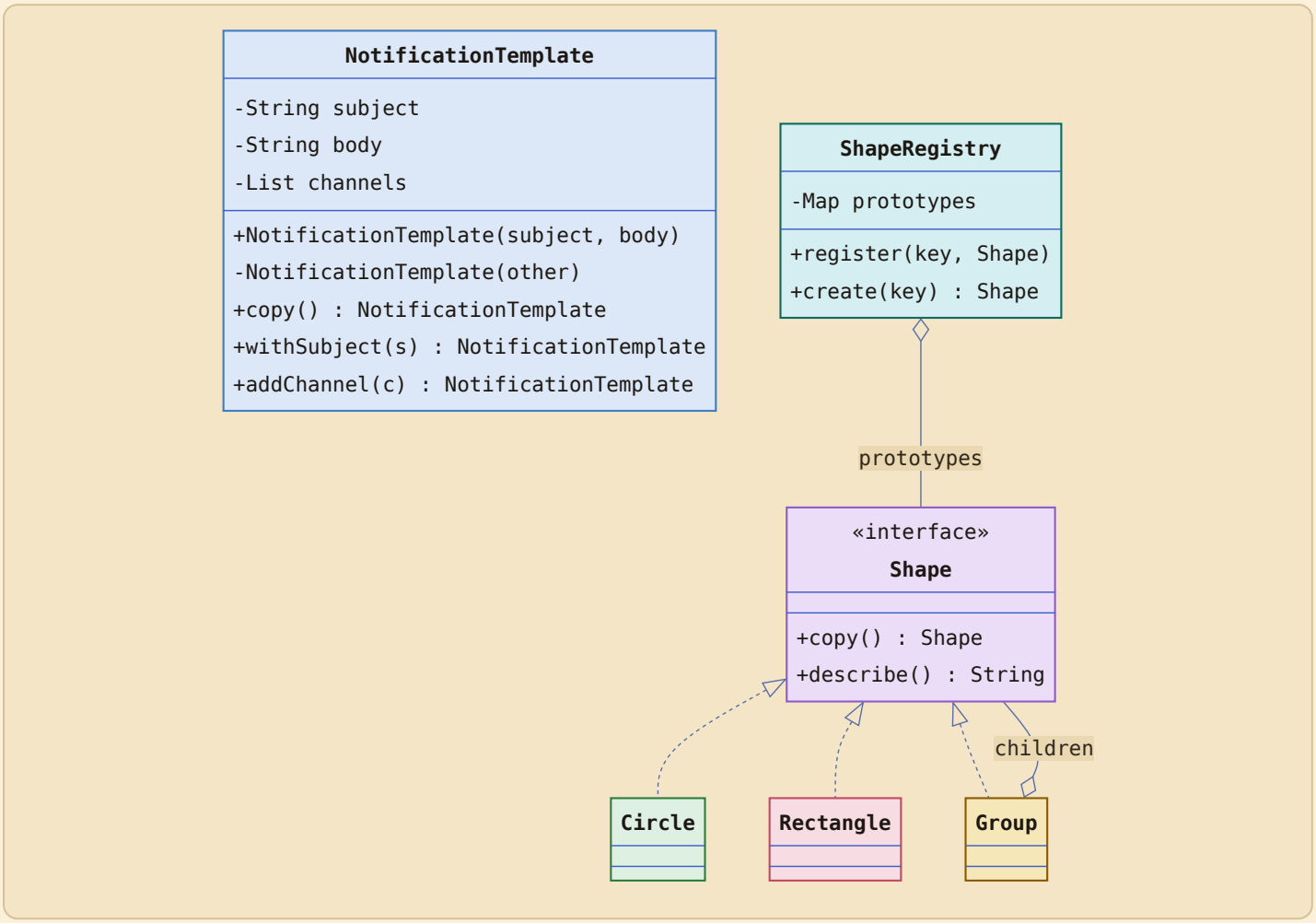


Figure — a shallow copy shares the channels list with the prototype, so adding "push" to the copy corrupts the prototype; a deep copy gets its own list (course diagram)

How it works

1. Two entrances: the **normal constructor** (expensive) and a **copy path** (none of it). 2. Implement the copy as a **private copy constructor**. 3. Per field: immutable → share; mutable → copy (`new ArrayList<>(other.channels)`), recursing for nested mutables. 4. Public `copy()`. 5. A **prototype interface** (`Shape { Shape copy(); }`) only when clients hold several unrelated types. 6. Optional **registry** that always hands out copies. (No separate creator class — the product copies itself.)



Shallow vs deep copy

	Shallow	Deep
Primitives, `String`, immutables	Copied / safely shared	Copied / safely shared
Mutable fields	**Shared**	**Duplicated**
Cost	Cheap	Proportional to the object graph
Risk	Editing the copy changes the original	Cycles; copying caches or resources
Java default	`Object.clone()`	You write it per field

With `this.channels = other.channels`, adding "push" to a copy adds it to the prototype — and **every later copy** inherits it. The fix is one line per mutable field. **Copy depth is a per-field design choice.** Related traps: a **new field not added to the copy** (silently dropped); copying something that **owns a resource** (both try to close it); **cycles** (naive recursive copy never ends — track visited objects).

Why Java's `Cloneable` is a trap

(*Effective Java* item 13.) `Cloneable` is a **marker with no methods**; `Object.clone()` is `protected`; it's **shallow**; it **skips constructors** (and their validation); deep-copying needs reassignment — **impossible for `final` fields**; it throws a **checked** exception. **Prefer a copy constructor or static `copyOf(other)`.** Exception: `array.clone()` for primitive arrays is fine.

Java implementation

```java
/** Prototype via a copy constructor: one expensive path, one cheap copy path. */
public final class NotificationTemplate {
    static int parses = 0;

    private String subject;
    private final String body;
    private final List<String> channels;

    /** The expensive path (imagine parsing a template file). */
    public NotificationTemplate(String subject, String body) {
        parses++;
        System.out.println("  Parsing template from disk... (expensive)");
        this.subject = subject;
        this.body = body;
        this.channels = new ArrayList<>(List.of("email"));
    }

    /** The copy path: no parsing. Deep-copies the mutable list. */
    private NotificationTemplate(NotificationTemplate other) {
        this.subject = other.subject;              // String is immutable -> sharing is safe
        this.body = other.body;
        this.channels = new ArrayList<>(other.channels);  // NEW list -> the copy owns its own
    }

    public NotificationTemplate copy() { return new NotificationTemplate(this); }
    // ... withSubject, addChannel, toString ...
}
```

Also in the package: `Group.copy()` (recursive deep copy of children), `Circle.copy()` with a covariant return, `ShapeRegistry.create()` returning `p.copy()` (never the original), and `CloneablePitfall.java` (`ShallowPlaylist` bug vs `DeepPlaylist`, whose `songs` can't be `final`).

Output: "Parsing template from disk" prints **once** for three templates, and the prototype keeps `channels=[email]` while copies add `push`/`sms`.

Full runnable code: code/lld/ch08/prototype/ (see the Full Edition).

Trade-offs

Pros	Cons
Costly setup paid once	**Copy code in every class** — drifts when fields are added
Create without naming a class	**Shallow vs deep per field** — wrong call shares state
New kinds at run time via registered prototypes	**Cyclic graphs** need care
Saved starting points (templates, layouts)	**Hazardous for identity** (keys, handles, connections); Java's `Cloneable` is flawed

Often confused with: Factory Method (fresh object); Builder (from parts); **Object Pool** (lends *existing* objects); **Spring** `prototype` **scope** (constructs anew — no cloning); JS prototypes (a language feature). JS spread is *shallow*; `structuredClone()` and Python `deepcopy` are deep.

When not to use it

Cheap construction; objects with **identity** (two rows sharing an ID is worse than a slow constructor); large graphs; immutable objects (just share them).

Common mistakes

Shallow-copying a mutable collection; trusting `super.clone()`; forgetting to update the copy constructor; handing out the registry's original.

Real-world examples

`ArrayList.clone()` / `HashMap.clone()` (**shallow**), copy constructors (`new ArrayList<>(c)`), `List.copyOf`; Unity `Instantiate(prefab)`; "Duplicate" in Docs/Figma; templates in Notion/Jira.

In the interview

> **30-second answer:** "Prototype creates objects by copying a configured instance via `copy()`. It saves expensive setup and — the deeper reason — lets me create an object when I only hold an interface. The main decision is shallow vs deep per field; a shared mutable list is the classic bug. In Java I'd write a copy constructor, not `Cloneable`, and never copy things with identity like a DB id or connection."

- *Q: Deep copy with cycles?* — A: An identity map original → copy; reuse the copy if already made.
- *Q: Serialization-based deep copy?* — A: Possible but slow, needs `Serializable`, bypasses constructors — last resort.

Self-check

- **Q:** Shallow vs deep? **A:** Shallow copies references; deep also copies the referenced mutable objects.
- **Q:** Why can't `DeepPlaylist.songs` be `final`? **A:** `clone()` must reassign it after `super.clone()`.
- **Q:** What should a registry return? **A:** A copy, never the original.

TL;DR

Copy a configured object — saves setup and works with only an interface. Decide shallow vs deep per field. Copy constructors, not `Cloneable`. Never copy identity.

8.14 Bonus — Object Pool

What it is

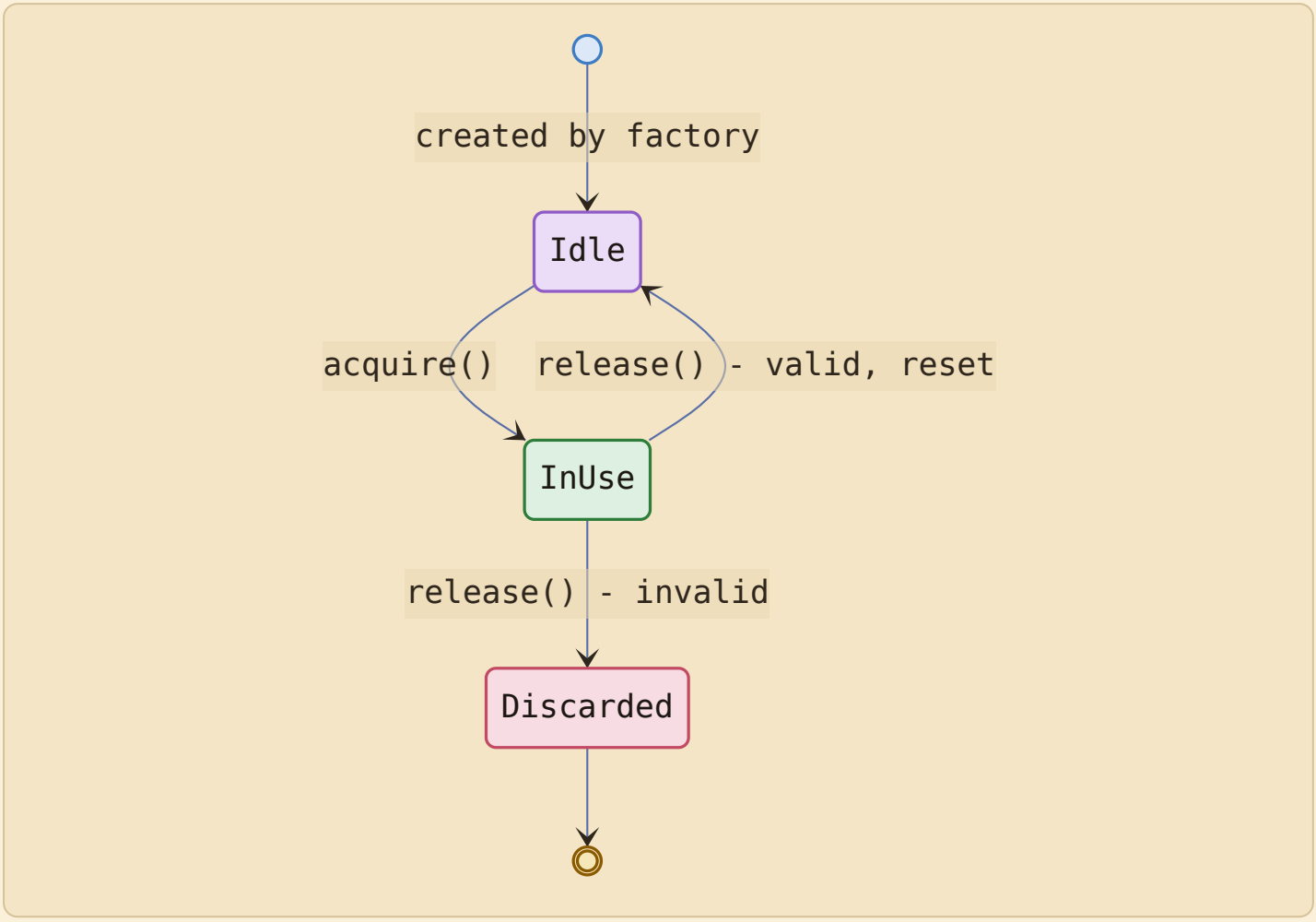
A **bounded set of reusable objects**: clients **acquire**, use, and **release** so others can reuse them. Not GoF, but a known creational pattern. Analogy: a bike-sharing dock — take, ride, return; if none, wait (or give up).

The problem it solves

Some objects are **expensive** (DB connection: TCP + TLS + auth) and/or **limited** (the DB caps connections). One per request is slow and can overwhelm the server. A pool pays creation a few times and **caps** the total.

How it works

1. An **idle queue**, a live count, and a cap (a `Semaphore` with one permit per lendable object).
2. **acquire()**: wait (with timeout) for a permit; take an idle object or create one; fail on timeout.
3. Client **always releases in** `finally` — a leak is lost forever.
4. **release()**: **validate** (discard broken ones), **reset** per-borrower state, return it; always free the permit.
5. Everything **thread-safe**.



Java implementation

```
// ... ObjectPool<T> fields: Semaphore permits (maxSize, fair), ConcurrentLinkedQueue<T> idle,
// ...   Supplier<T> factory, Consumer<T> reset, Predicate<T> isValid, AtomicInteger created
    /** Borrow an object, waiting up to the timeout if all are in use. */
    public T acquire(long timeout, TimeUnit unit) throws InterruptedException {
        if (!permits.tryAcquire(timeout, unit)) {
            throw new IllegalStateException("pool exhausted: timed out after " + timeout + " " + unit);
        }
        // Holding a permit guarantees we are within the cap: reuse an idle object or create one.
        T obj = idle.poll();
        if (obj != null) return obj;
        try {
            obj = factory.get();
            created.incrementAndGet();
            return obj;
        } catch (RuntimeException e) {
            permits.release();                                  // creation failed: give the permit back
            throw e;
        }
    }

    /** Give an object back. Broken objects are discarded so a fresh one can be created later. */
    public void release(T obj) {
        try {
            if (isValid.test(obj)) {
                reset.accept(obj);
                idle.offer(obj);
            } else {
                created.decrementAndGet();                      // discard it; its permit still comes back
            }
        } finally {
            permits.release();                                  // wakes one waiting borrower, if any
        }
    }
```

- The `Semaphore` caps borrowed objects however many threads race; holding a permit means "reuse idle, else safe to create".
- `release()` frees the permit **even when discarding**, so a waiter can create a replacement; a failed factory call also returns its permit.
- Handing objects through a concurrent queue + semaphore makes the previous borrower's writes visible to the next.
- `Main`: 10 tasks on 6 threads with a pool of 3 → `connections ever created: 3`; exhaustion prints `pool exhausted: timed out after 100 MILLISECONDS`.

Full runnable code: code/lld/ch08/objectpool/ (see the Full Edition).

Trade-offs

Pros	Cons
Avoids repeated expensive creation	Complexity: sizing, validation, timeouts, leak detection
Caps resource usage (protects the DB)	Leaks shrink the pool until it deadlocks
Predictable latency after warm-up	Incomplete reset leaks one user's data to the next
	Slower than `new` for cheap objects

When not to use it

Cheap objects (JVM allocation is faster); **virtual threads** (Java 21 — create one per task); when a **library** exists (HikariCP, Commons Pool 2).

Common mistakes

Not releasing in `finally`; not resetting state; unbounded waits; **double release** (here it would queue one object twice *and* add a permit, silently raising the cap — real pools track borrowed objects).

Real-world examples

HikariCP (Spring Boot default), DBCP, c3p0; `ThreadPoolExecutor`; Commons Pool 2 `GenericObjectPool` (Jedis); Netty `PooledByteBufAllocator`. *Not* pools: the `String` intern pool and `Integer` cache (-128..127) share immutables — that's **Flyweight** (Ch. 9).

In the interview

> **30-second answer:** "An object pool reuses a bounded set of expensive objects like DB connections. acquire takes an idle one, creates one if under the cap, or waits with a timeout; release validates, resets and returns it. It must be thread-safe and clients release in finally. For cheap objects it's counter-productive; in production I'd use HikariCP."

- *Q: Sizing?* — A: From measured concurrency and downstream limits; small pools (≈ core-count scale) often win.
- *Q: Leaks?* — A: Record borrower and time; log long holds (HikariCP `leakDetectionThreshold`).

Self-check

- **Q:** Pool vs Flyweight? **A:** Pool lends mutable objects exclusively and takes them back; Flyweight shares immutable objects with everyone at once.
- **Q:** Why validate on release? **A:** So dead objects aren't handed to the next borrower.

TL;DR

Borrow → use → return (in `finally`). Bounded, thread-safe, validated, reset. Only for expensive/limited resources; use a mature library.

8.15 Creational patterns compared

At a glance

Pattern	Distinctive feature	Use when	JDK / Spring
Singleton	One instance	One shared resource	`Runtime.getRuntime()`, singleton beans
Simple Factory	One method maps type → class	Small, stable, config-chosen set	`Calendar.getInstance()`, `EnumSet.noneOf`, `@Bean`
Factory Method	Subclasses choose via overridable method	Shared procedure, varying product	`Collection.iterator()`, `FactoryBean`
Abstract Factory	Families without concrete classes	Products must match	`DocumentBuilderFactory`, JDBC `Connection`, profiles
Builder	Construction separate from representation	Optional fields, validation, immutability	`StringBuilder`, `HttpClient.newBuilder()`
Prototype	Copy an existing object	Expensive setup, or only interface known	Copy constructors, `ArrayList.clone()`, Unity
Object Pool	Reuse a bounded set	Expensive/limited resources	HikariCP, `ThreadPoolExecutor`

Pick by the question you are asking

You are asking…	Pattern
"Only one?"	Singleton (or DI singleton scope)
"Avoid scattered `new ConcreteClass()`?"	Simple Factory
"Let subclasses/plug-ins decide?"	Factory Method
"Make sure these always match?"	Abstract Factory
"Make a 10-argument constructor readable and safe?"	Builder
"Another one of *this*, cheaply or without knowing its class?"	Prototype
"Stop creating and destroying this expensive thing?"	Object Pool

How they combine

- **Abstract Factory** is often built from **Factory Methods**, may use **Prototypes**, and is often a **Singleton** (passing it in is better).
- A **Builder** can use a **Factory** to pick parts; a **Director** can be a stateless shared service.
- Designs often *start* with Factory Method and *evolve* toward Abstract Factory, Prototype or Builder.

8.16 Chapter self-check

1. **Q:** The three GoF categories and counts? **A:** Creational 5, Structural 7, Behavioral 11 = 23.
2. **Q:** The two GoF central principles? **A:** Program to an interface; favour composition over inheritance.
3. **Q:** Why must the DCL field be `volatile`? **A:** So no thread sees a reference to a not-yet-constructed object (happens-before).
4. **Q:** Factory Method vs Abstract Factory? **A:** One product by inheritance vs a matching family by composition.
5. **Q:** How many patterns in a 45-minute design? **A:** One or two — and name the one you rejected.

8.17 Chapter TL;DR

- **Patterns** = named, proven templates with trade-offs; GoF (1994) catalogued 23 in three categories.
- **Pick from the signal**, one or two per design; state cost and alternative.
- **Singleton** — one per JVM; eager / holder / enum; DCL needs `volatile`; prefer DI.
- **Simple Factory** — one `create(type)`; idiom, not GoF.
- **Factory Method** — base-class procedure + subclass-chosen product.
- **Abstract Factory** — one factory, one matching family, chosen at the composition root.
- **Builder** — named steps, one validating `build()`, immutable product.
- **Prototype** — copy a configured object; shallow vs deep; avoid `Cloneable`.
- **Object Pool** — borrow/return expensive objects; use a real library.